

Problem-Solving & Data



Ratios and proportions

- 1 A recipe calls for 3 cups of flour for every 2 cups of sugar. If a baker uses 9 cups of flour, how many cups of sugar are needed?
 - 2 On a map, 1 inch represents 25 miles. If two cities are 7 inches apart on the map, what is the actual distance between them?
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Percent problems

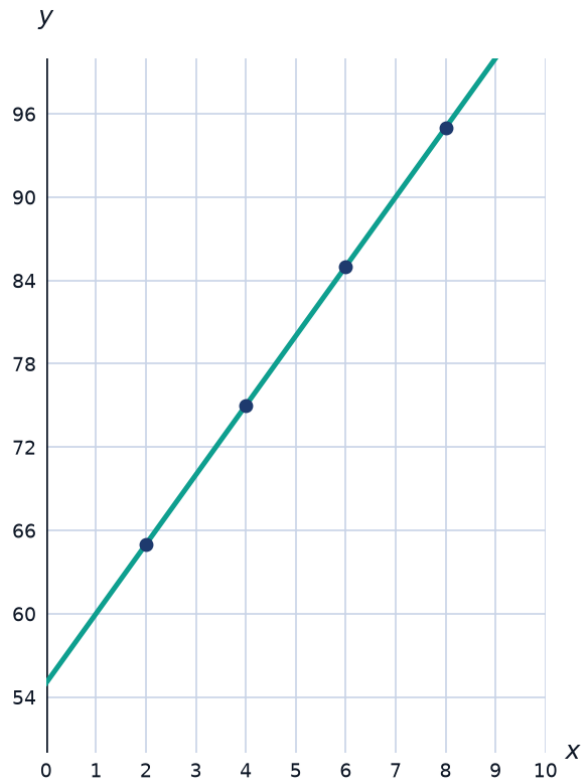
- 3 A shirt originally priced at 40 dollars is on sale for 25 percent off. Find the sale price.
 - 4 A population increased from 4000 to 4600. Find the percent increase.
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Unit conversions and rates

- 5 A car travels at 60 miles per hour. How many miles does it travel in 45 minutes?
 - 6 A factory produces 480 units in an 8-hour shift. Find the production rate in units per minute.
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Reading and interpreting scatterplots and tables

- 7 A table shows study hours and test scores: (2,65), (4,75), (6,85), (8,95). Describe the relationship between study hours and test score, and predict the score for 5 hours of study.

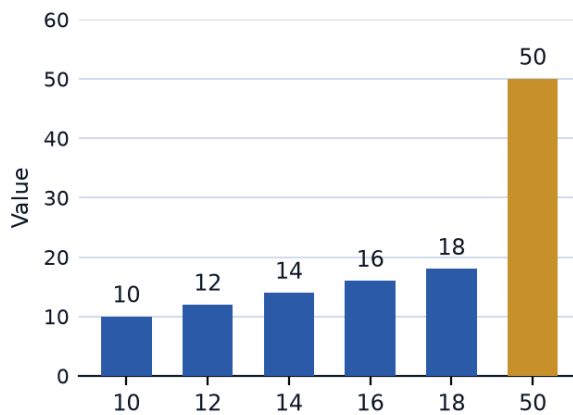


Measures of center: mean, median, and mode

- 8 Find the mean, median, and mode of the data set: 4, 7, 7, 9, 12, 15.
- 9 A data set has mean 20 and 5 values. If four of the values are 15, 18, 22, 25, find the fifth value.

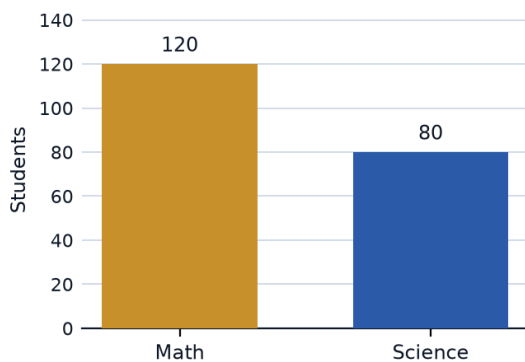
Measures of spread and the effect of outliers

- 10 Find the range and explain the effect on the mean if an outlier of 50 is added to the data set 10, 12, 14, 16, 18.

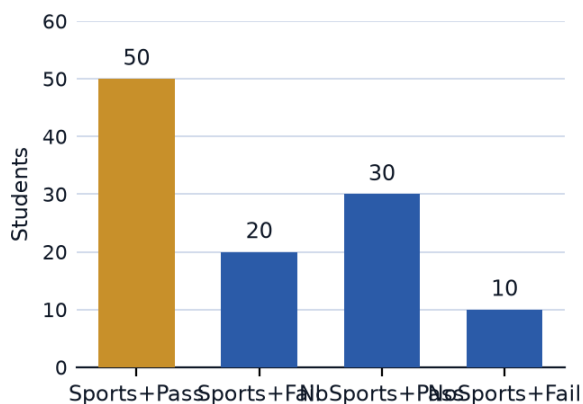


Probability from data tables

- 11 A survey of 200 students found 120 prefer math and 80 prefer science. If a student is selected at random, find the probability the student prefers math.



- 12 A two-way table shows: 50 students play sports and pass an exam; 20 play sports and fail; 30 don't play sports and pass; 10 don't play sports and fail. Find the probability a randomly selected student plays sports, given the student passed the exam.



Population, sample, and margin of error

- 13** A poll of 500 randomly selected voters found that 55% support a proposal, with a margin of error of 4%. Interpret this result and state the range of plausible values for the true population percentage.

Weighted averages

- 14** A student's final grade is weighted: homework counts 20%, tests count 50%, and the final exam counts 30%. If the student scores 90 on homework, 80 on tests, and 85 on the final exam, find the weighted final grade.
- 15** A store mixes 3 pounds of coffee costing 8 dollars per pound with 2 pounds of coffee costing 13 dollars per pound. Find the price per pound of the mixture.

Multi-step unit conversion problems

- 16** A machine produces 15 items per minute. Find the number of items produced in 3.5 hours.
- 17** A car uses fuel at a rate of 28 miles per gallon. If gas costs 3.50 dollars per gallon, find the fuel cost for a 420-mile trip.

Percent change applied twice (successive changes)

- 18** A stock price increases by 20% one year, then decreases by 20% the next year. If the original price was 100 dollars, find the final price, and explain why it is not back to 100 dollars.

Density and rate-based problem solving

- 19** A population density is 150 people per square mile. If a city has an area of 24 square miles, find the total population.
- 20** Three workers can complete a job in 12 hours together. If they work at the same combined rate, how many hours would it take 2 workers with the same total capacity to complete a job that is 1.5 times as large?

Multiple choice — easy 15

- 21** A recipe calls for 4 cups of flour for every 9 cups of sugar. If a baker uses 32 cups of flour, how many cups of sugar are needed?
- a. 80
 - b. 32
 - c. 64
 - d. 72
- 22** A recipe calls for 7 cups of rice for every 2 cups of beans. If a baker uses 77 cups of rice, how many cups of beans are needed?
- a. 77
 - b. 33
 - c. 11
 - d. 22
- 23** A recipe calls for 5 cups of water for every 2 cups of concentrate. If a baker uses 20 cups of water, how many cups of concentrate are needed?
- a. 4
 - b. 12
 - c. 20
 - d. 8
- 24** A recipe calls for 5 cups of water for every 4 cups of concentrate. If a baker uses 15 cups of water, how many cups of concentrate are needed?
- a. 8
 - b. 15
 - c. 12
 - d. 9

- 25** A recipe calls for 9 cups of red paint for every 3 cups of blue paint. If a baker uses 72 cups of red paint, how many cups of blue paint are needed?
- a. 32
 - b. 16
 - c. 24
 - d. 72
- 26** A recipe calls for 8 cups of red paint for every 3 cups of blue paint. If a baker uses 48 cups of red paint, how many cups of blue paint are needed?
- a. 12
 - b. 48
 - c. 18
 - d. 24
- 27** A recipe calls for 5 cups of rice for every 9 cups of beans. If a baker uses 35 cups of rice, how many cups of beans are needed?
- a. 63
 - b. 56
 - c. 35
 - d. 70
- 28** A recipe calls for 8 cups of cement for every 5 cups of sand. If a baker uses 96 cups of cement, how many cups of sand are needed?
- a. 60
 - b. 96
 - c. 72
 - d. 48

- 29** A recipe calls for 6 cups of flour for every 2 cups of sugar. If a baker uses 36 cups of flour, how many cups of sugar are needed?
- a. 18
 - b. 6
 - c. 12
 - d. 36
- 30** A recipe calls for 7 cups of red paint for every 4 cups of blue paint. If a baker uses 28 cups of red paint, how many cups of blue paint are needed?
- a. 28
 - b. 16
 - c. 12
 - d. 20
- 31** A recipe calls for 7 cups of flour for every 5 cups of sugar. If a baker uses 56 cups of flour, how many cups of sugar are needed?
- a. 32
 - b. 56
 - c. 48
 - d. 40
- 32** A recipe calls for 8 cups of rice for every 5 cups of beans. If a baker uses 40 cups of rice, how many cups of beans are needed?
- a. 25
 - b. 40
 - c. 20
 - d. 30

- 33** A recipe calls for 9 cups of rice for every 5 cups of beans. If a baker uses 27 cups of rice, how many cups of beans are needed?
- a. 15
 - b. 12
 - c. 18
 - d. 27
- 34** A recipe calls for 8 cups of water for every 4 cups of concentrate. If a baker uses 32 cups of water, how many cups of concentrate are needed?
- a. 32
 - b. 20
 - c. 16
 - d. 12
- 35** A recipe calls for 5 cups of red paint for every 2 cups of blue paint. If a baker uses 40 cups of red paint, how many cups of blue paint are needed?
- a. 40
 - b. 16
 - c. 24
 - d. 8

Multiple choice — medium 15

- 36 A desk originally priced at 70 dollars is on sale for 40 percent off. Find the sale price.
- a. 42
 - b. 47
 - c. 28
 - d. 37
- 37 A laptop originally priced at 105 dollars is on sale for 20 percent off. Find the sale price.
- a. 84
 - b. 89
 - c. 21
 - d. 79
- 38 A bicycle originally priced at 180 dollars is on sale for 10 percent off. Find the sale price.
- a. 157
 - b. 18
 - c. 167
 - d. 162
- 39 A watch originally priced at 200 dollars is on sale for 25 percent off. Find the sale price.
- a. 50
 - b. 150
 - c. 155
 - d. 145

- 40 A bicycle originally priced at 160 dollars is on sale for 35 percent off. Find the sale price.
- a. 99
 - b. 104
 - c. 56
 - d. 109
- 41 A laptop originally priced at 20 dollars is on sale for 30 percent off. Find the sale price.
- a. 6
 - b. 19
 - c. 9
 - d. 14
- 42 A television originally priced at 200 dollars is on sale for 15 percent off. Find the sale price.
- a. 30
 - b. 175
 - c. 165
 - d. 170
- 43 A laptop originally priced at 185 dollars is on sale for 20 percent off. Find the sale price.
- a. 148
 - b. 37
 - c. 143
 - d. 153

- 44 A laptop originally priced at 110 dollars is on sale for 20 percent off. Find the sale price.
- a. 88
 - b. 83
 - c. 93
 - d. 22
- 45 A bicycle originally priced at 75 dollars is on sale for 40 percent off. Find the sale price.
- a. 45
 - b. 50
 - c. 30
 - d. 40
- 46 A backpack originally priced at 120 dollars is on sale for 5 percent off. Find the sale price.
- a. 114
 - b. 119
 - c. 109
 - d. 6
- 47 A desk originally priced at 130 dollars is on sale for 10 percent off. Find the sale price.
- a. 13
 - b. 122
 - c. 117
 - d. 112

- 48 A backpack originally priced at 30 dollars is on sale for 40 percent off. Find the sale price.
- a. 12
 - b. 18
 - c. 13
 - d. 23
- 49 A television originally priced at 80 dollars is on sale for 35 percent off. Find the sale price.
- a. 57
 - b. 47
 - c. 52
 - d. 28
- 50 A laptop originally priced at 40 dollars is on sale for 15 percent off. Find the sale price.
- a. 6
 - b. 29
 - c. 34
 - d. 39

Multiple choice — hard 15

- 51 A student's final grade is weighted: homework counts 30%, tests count 35%, and the final exam counts 35%. If the student scores 78 on homework, 65 on tests, and 90 on the final exam, find the weighted final grade.
- a. 77.67
 - b. 74.65
 - c. 77.65
 - d. 80.65

- 52** A student's final grade is weighted: homework counts 15%, tests count 35%, and the final exam counts 50%. If the student scores 80 on homework, 93 on tests, and 73 on the final exam, find the weighted final grade.
- a. 84.05
 - b. 82.00
 - c. 81.05
 - d. 78.05
- 53** A student's final grade is weighted: homework counts 20%, tests count 50%, and the final exam counts 30%. If the student scores 80 on homework, 98 on tests, and 66 on the final exam, find the weighted final grade.
- a. 81.33
 - b. 84.80
 - c. 87.80
 - d. 81.80
- 54** A student's final grade is weighted: homework counts 15%, tests count 35%, and the final exam counts 50%. If the student scores 86 on homework, 82 on tests, and 91 on the final exam, find the weighted final grade.
- a. 84.10
 - b. 87.10
 - c. 90.10
 - d. 86.33

- 55** A student's final grade is weighted: homework counts 10%, tests count 35%, and the final exam counts 55%. If the student scores 72 on homework, 87 on tests, and 94 on the final exam, find the weighted final grade.
- a. 92.35
 - b. 86.35
 - c. 84.33
 - d. 89.35
- 56** A two-way table shows: 32 students take the bus and arrive on time; 19 take the bus but do not arrive on time; 33 do not take the bus but do arrive on time; 33 do neither. Find the probability a randomly selected student takes the bus, given that the student arrives on time.
- a. 0.49
 - b. 0.39
 - c. 0.59
 - d. 0.63
- 57** A two-way table shows: 47 students own a car and commute over 30 minutes; 38 own a car but do not commute over 30 minutes; 39 do not own a car but do commute over 30 minutes; 17 do neither. Find the probability a randomly selected student owns a car, given that the student commutes over 30 minutes.
- a. 0.65
 - b. 0.55
 - c. 0.45
 - d. 0.60

- 58** A two-way table shows: 31 students like coffee and like tea; 25 like coffee but do not like tea; 36 do not like coffee but do like tea; 34 do neither. Find the probability a randomly selected student likes coffee, given that the student likes tea.
- a. 0.36
 - b. 0.56
 - c. 0.55
 - d. 0.46
- 59** A two-way table shows: 31 students take the bus and arrive on time; 30 take the bus but do not arrive on time; 13 do not take the bus but do arrive on time; 34 do neither. Find the probability a randomly selected student takes the bus, given that the student arrives on time.
- a. 0.70
 - b. 0.51
 - c. 0.80
 - d. 0.60
- 60** A two-way table shows: 26 students like coffee and like tea; 36 like coffee but do not like tea; 5 do not like coffee but do like tea; 36 do neither. Find the probability a randomly selected student likes coffee, given that the student likes tea.
- a. 0.42
 - b. 0.74
 - c. 0.94
 - d. 0.84

- 61 A price of 200 dollars increases by 25%, then the new price increases by 25%. Find the final price.
- a. 200.00
 - b. 312.50
 - c. 317.50
 - d. 307.50
- 62 A price of 120 dollars increases by 10%, then the new price increases by 10%. Find the final price.
- a. 150.20
 - b. 140.20
 - c. 120.00
 - d. 145.20
- 63 A price of 120 dollars decreases by 30%, then the new price increases by 25%. Find the final price.
- a. 100.00
 - b. 105.00
 - c. 110.00
 - d. 120.00
- 64 A price of 120 dollars increases by 25%, then the new price decreases by 30%. Find the final price.
- a. 110.00
 - b. 100.00
 - c. 120.00
 - d. 105.00

- 65 A price of 120 dollars decreases by 10%, then the new price increases by 15%. Find the final price.
- a. 120.00
 - b. 119.20
 - c. 124.20
 - d. 129.20